rocket engines, but wasting a free resource did not make sense, especially when the neighbouring Aralvaymozhi Pass and Aryankavu Pass were becoming success stories in wind energy tapping. Anyway, I didn't push much. I choose my battles.

With my immediate boss Muthunayagam, I believed, some battles have to be openly fought. When I organised a *bhumi puja* for the 'groundbreaking' ceremony at Mahendragiri, Muthunayagam was not happy.

'Why a Hindu ritual?' he said.

'Why do you see it as a Hindu ritual?' I replied. 'It is just a prayer to Mother Earth. We may bring a Christian priest and a mullah, too. Let there be all kinds of rituals. A bhumi puja is a routine in virtually every big project in India, including government ones. And, mind you, I have arranged it with my money, not ISRO's.'

The puja was conducted.

The pace at which infrastructure for the Mahendragiri integrated test complex was put together displayed the efficiency of not just ISRO and many government arms, but also the inherent problems of bureaucracy. While the land acquisition and construction of the facility structures happened in record time, acquiring the first computer was a fifteen-month struggle. Overcoming all such delays, the integrated test facility was up and running about three years after the land acquisition was completed. And the first test of a gas generator was done here on 31 December 1985.

By then our first Vikas engine was successfully tested at SEP, which had two test stands, one for single engines and another for four-engine clusters (now used by Arianne). We set up the principal test stand at Mahendragiri as a virtual replica of the SEP facility, an unofficial technology transfer we managed simply through observation and copying for five years. The principal test stand was approved by the ISRO board as a single engine test facility, but just as the ground work was beginning I had a thought, 'Why not build a principal stand for testing a four-engine cluster like Arianne?'

It was not a big deal. The square foundation for the single-engine test stand, as approved, was 4m x 4m. It has to be increased to 5m x 5m. We